

CALIBRATION KIT

Model ST-8770S40 DC-40GHz Type 2.92mm

SOLT Calibration Kit



INSTRUMENT LIST

Item No.	Instrument	Model Number	Quantity
1	Open (Male)	ST-8770S40-KM	1
2	Open (Female)	ST-8770S40-KF	1
3	Short (Male)	ST-8770S40-DM	1
4	Short (Female)	ST-8770S40-DF	1
5	Load (Male)	ST-8770S40-LM	1
6	Load (Female)	ST-8770S40-LF	1
7	Storage Box 275*200*810mm		1
Optional accessories			
1	Adaptor: 2.92(M)-2.92(M)	Optional	
2	Adaptor: 2.92(F)-2.92(F)	Optional	
3	Adaptor: 2.92(M)-2.92(F)	Optional	
4	Solid Wrench: Opening size 7mm	Optional	
5	Solid Wrench: Opening size 8mm	Optional	
6	Torque wrench: 8mm(5/16in), 0.9Nm	Optional	

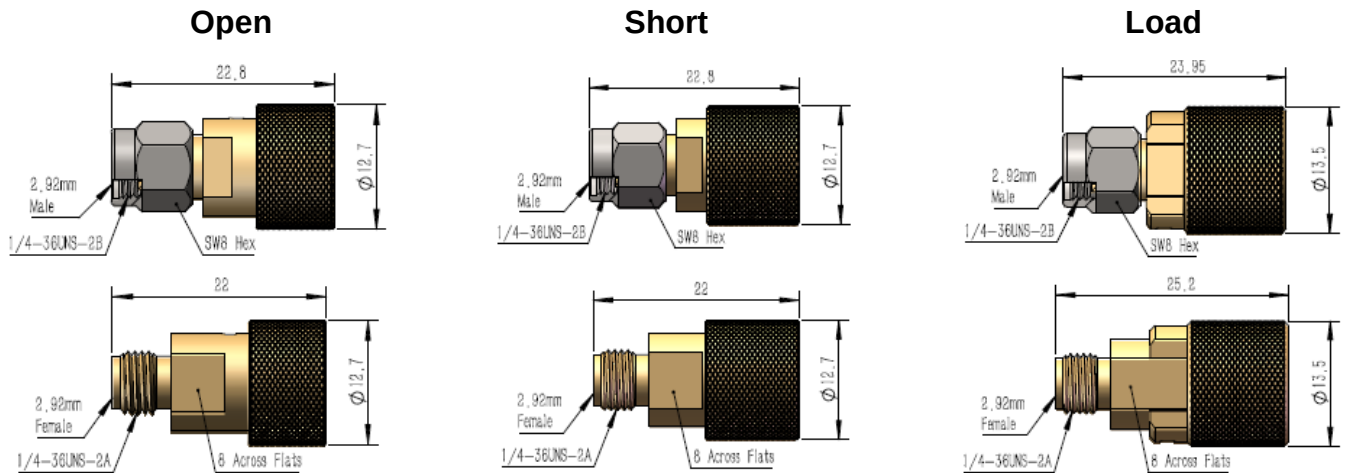
SPECIFICATIONS

Instrument	Model Number	Frequency Range	Specifications
Open (Male)	ST-8770S40-KM	DC-40GHz	Phase deviation: $\leq \pm 6^\circ$ *
Open (Female)	ST-8770S40-KF		Phase deviation: $\leq \pm 6^\circ$ *
Short (Male)	ST-8770S40-DM		Phase deviation: $\leq \pm 4.5^\circ$ *
Short (Female)	ST-8770S40-DF		Phase deviation: $\leq \pm 4.5^\circ$ *
Load (Male)	ST-8770S40-LM		VSWR: ≤ 1.052
Load (Female)	ST-8770S40-LF		VSWR: ≤ 1.052

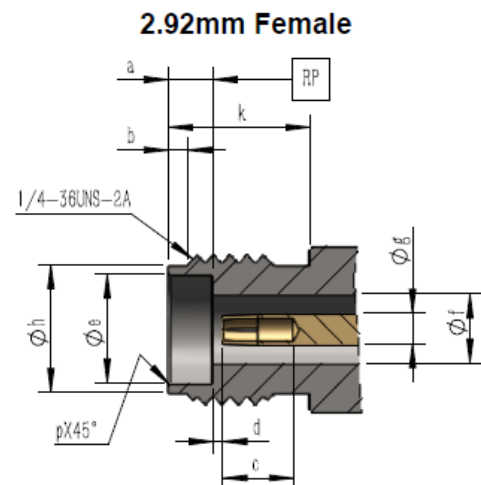
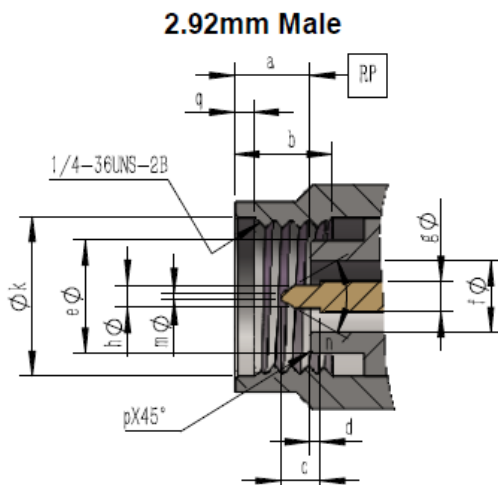
SURFACE TREATMENT

Instrument	Inner Conductor	Outer Conductor
Open	Beryllium bronze, gold-plated	Beryllium bronze, gold-plated
Short	Beryllium bronze, gold-plated	Beryllium bronze, gold-plated
Load	Beryllium bronze, gold-plated	Beryllium bronze, gold-plated

OUTDRAWING:



INTERFACE STANDARD:



	mm		in	
	Min.	Max.	Min.	Max.
a	2.36	3.56	0.093	0.135
b	3.43	4.01	0.135	0.158
c	1.52	1.62	0.060	0.064
d	0.000	0.030	0.000	0.0012
e	4.512	4.592	0.1780	0.1808
f	2.91	2.93	0.1145	0.1153
g	1.27Ref		0.050Ref	
h	0.906	0.922	0.0357	0.0363
k	6.48	6.73	0.255	0.265
m	0.20	0.30	0.008	0.012
n	55°	64°	56°	64°
p	0.1		0.004	
q	0.38	1.14	0.015	0.045

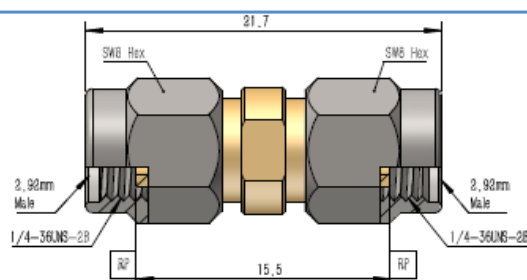
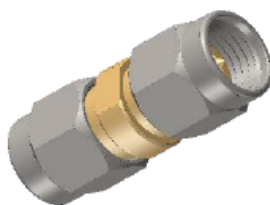
	mm		in	
	Min.	Max.	Min.	Max.
a	1.88	1.98	0.074	0.078
b	0.38	1.14	0.015	0.045
c	2.65	---	0.104	---
d	0.00	0.030	0.0000	0.0012
e	4.60	4.65	0.181	0.183
f	2.90	2.95	0.114	0.116
g	1.27 Ref.		0.050 Ref.	
h	5.28	5.46	0.208	0.215
k	5.54	---	0.218	---
p	0.1		0.004	

CALIBRATION KIT

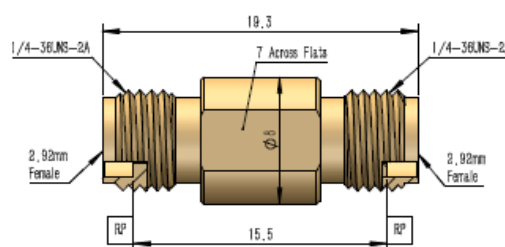
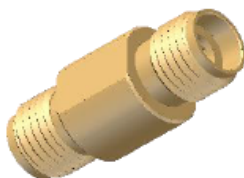
ADAPTOR:

Instrument	Frequency Range	Impedance	VSWR	Cycle Life	Temp Range
2.92(M)-2.92(M)	DC-40GHz	50Ω	≤1.12	> 2000 times	-55~+85℃
2.92(F)-2.92(F)			≤1.12		
2.92(M)-2.92(F)			≤1.12		
Material and Surface Treatment					
Outer Conductor		Inner Conductor		Insulator	Threaded sleeve
Beryllium bronze gold-plated		Beryllium bronze gold-plated		PEEK	Stainless steel SUS303 passivated

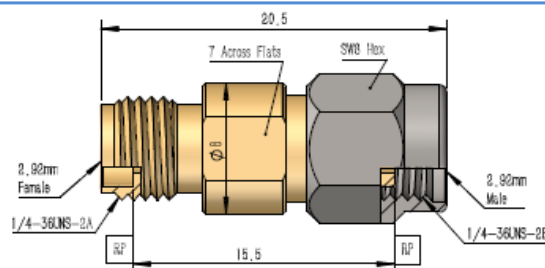
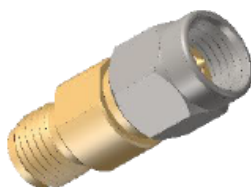
2.92mm (m) to 2.92mm (m)



2.92mm (f) to 2.92mm (f)



2.92mm (m) to 2.92mm (f)

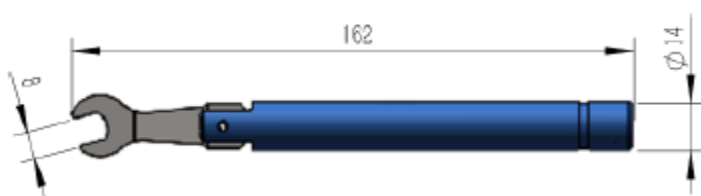


Solid Wrench



Aluminum alloy anodizing

Torque wrench:



Opening size: 8mm (5/16 inch)

Torque setting: 0.9Nm (8 inch-lbs)

Wrench handle: Aluminum, anodized

Wrench head: Tool steel, nickel plated



CALIBRATION PARAMETERS:

OPEN		Male	Female
Z0		50Ω	50Ω
Fringing Capacitances	C0 F(e-15)	44.1578	42.9684
	C1 F(e-27)/Hz	71.4204	729.336
	C2 F(e-36)/Hz ²	-0.1716	-31.7551
	C3 F(e-45)/Hz ³	0.2048	0.6628
Offset Delay (ps)		14.8487	14.8487
Offset length (mm)		4.452	4.452
Offset Loss	Gohm/Sec	3.39	3.4628
	dB/(GHz ^{1/2})	0.0087	0.0089
	dB/mm@1GHz	9.8e-4	10.0e-4

SHORT		Male	Female
Z0		50Ω	50Ω
Short Inductance	L0 H(e-12)	8.7413	-11.2831
	L1 H(e-24)/Hz	-1036.9	1910.57
	L2 H(e-33)/Hz ²	41.5223	-85.3145
	L3 H(e-42)/Hz ³	-0.5055	1.0846
Offset Delay (ps)		16.6963	16.6963
Offset length (mm)		5	5
Offset Loss	Gohm/Sec	2.5639	2.0059
	dB/(GHz ^{1/2})	0.0074	0.0058
	dB/mm@1GHz	7.4e-4	5.8e-4

LOAD	
Z0	50Ω
Offset Delay	0ps
Offset length	0mm
Offset Loss	Gohm/Sec 0dB/(GHz ^{1/2}) 0dB/mm

LOAD	
Z0	50Ω
Connector Port1	2.92mm female
Connector Port2	2.92mm male
Offset Delay	0ps
Offset length	0mm
Offset Loss	Gohm/Sec 0dB/(GHz ^{1/2}) 0dB/mm